ModelIt!

Student Activity Pack

Can a River Change Direction?

How Flowing Water Reshapes the Land Over Time

Grade 05L2 | NGSS 5-ESS2-1, 5-ESS2-2 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever seen a place where water wore away the ground? What did it look like?

2 Do you think rivers have always been in the same place? Why or why not?

3 Draw what you think happens to a riverbank when water flows past it very quickly.

4 What do you think 'erosion' means? Where have you seen it happen?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Flow Speed | Soil Softness | Erosion Rate | Sediment Transport | Channel Shape Change

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fast River, Soft Soil

Maximum flow speed, very soft soil

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Slow River, Soft Soil

Low flow speed, very soft soil

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Fast River, Hard Rock

Maximum flow speed, very hard rock

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Vegetation Cover — Plant roots hold soil together, reducing erosion even when water flows fast

Rainfall Intensity — Heavy rain adds more water to the river, increasing flow speed and erosion

River Slope — Steeper river channels cause water to flow faster, increasing erosive power

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can a river actually change the direction it flows? What forces make that happen? Use evidence from your simulation to support your answers.

1. **Why do rivers have curves instead of flowing in straight lines?**

2. **How is a river like a living thing that changes and grows over time?**

3. **What surprised you most about how quickly soft soil erodes?**

4. **If you could add one more component to this model, what would it be and why?**

5. **How does understanding erosion help communities plan where to build?**

8 STEM Challenge: Design a Riverbank Protection Plan

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YOUR ENGINEERING MISSION

Design a plan to protect a community's riverbank from erosion without harming the river ecosystem.

THE SCENARIO

A riverside community is losing land to erosion. Every year, the river eats away more of the bank, threatening homes and roads. Your team must design a protection plan that slows erosion while keeping the river healthy for fish and wildlife.

Engineering Design Thinking

1. WHAT MATERIALS OR NATURAL FEATURES WILL YOU USE TO SLOW EROSION?
2. HOW WILL YOU PROTECT THE BANK WITHOUT SPEEDING UP EROSION ELSEWHERE?
3. HOW WILL YOU ENSURE FISH AND WILDLIFE CAN STILL USE THE RIVER?
4. HOW WILL YOU MEASURE WHETHER YOUR PROTECTION PLAN IS WORKING OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Fluvial Geomorphologists and River Engineers study how rivers shape the land and design solutions to protect communities from erosion and flooding. They earn $65,000-$120,000/year.